

Effects of Homocysteine-Lowering With Folic Acid Plus Vitamin B₁₂ vs Placebo on Mortality and Major Morbidity in Myocardial Infarction Survivors

A Randomized Trial

Study of the Effectiveness of Additional Reductions in Cholesterol and Homocysteine (SEARCH) Collaborative Group*

OBSERVATIONAL STUDIES have consistently indicated that patients with occlusive vascular disease have higher blood levels of homocysteine than do controls, and these differences precede the onset of disease and are independent of other risk factors.¹⁻³ A meta-analysis of prospective studies indicated that, after adjustment for known risk factors, a 25% lower usual homocysteine concentration was associated with 11% lower risk of coronary heart disease and 19% lower risk of stroke.³ Daily supplementation with folic acid typically lowers homocysteine levels by about 25%, and the addition of vitamin B₁₂ lowers it by a further 7%.^{4,5} Folic acid is inexpensive, so there is considerable interest in the possibility of reducing the incidence of occlusive vascular disease with folic acid supplementation.

Seven large-scale randomized trials have previously reported on the effects on occlusive vascular disease of lowering homocysteine with folic acid-based vitamin supplements, but none of those trials has shown definite protective effects.⁶⁻¹² It has been unclear whether these generally unpromising results are due to insufficient numbers of particular cardiovascular events, too short a duration of treatment, at-

Context Blood homocysteine levels are positively associated with cardiovascular disease, but it is uncertain whether the association is causal.

Objective To assess the effects of reducing homocysteine levels with folic acid and vitamin B₁₂ on vascular and nonvascular outcomes.

Design, Setting, and Patients Double-blind randomized controlled trial of 12 064 survivors of myocardial infarction in secondary care hospitals in the United Kingdom between 1998 and 2008.

Interventions 2 mg folic acid plus 1 mg vitamin B₁₂ daily vs matching placebo.

Main Outcome Measures First major vascular event, defined as major coronary event (coronary death, myocardial infarction, or coronary revascularization), fatal or nonfatal stroke, or noncoronary revascularization.

Results Allocation to the study vitamins reduced homocysteine by a mean of 3.8 μmol/L (28%). During 6.7 years of follow-up, major vascular events occurred in 1537 of 6033 participants (25.5%) allocated folic acid plus vitamin B₁₂ vs 1493 of 6031 participants (24.8%) allocated placebo (risk ratio [RR], 1.04; 95% confidence interval [CI], 0.97-1.12; *P* = .28). There were no apparent effects on major coronary events (vitamins, 1229 [20.4%], vs placebo, 1185 [19.6%]; RR, 1.05; 95% CI, 0.97-1.13), stroke (vitamins, 269 [4.5%], vs placebo, 265 [4.4%]; RR, 1.02; 95% CI, 0.86-1.21), or noncoronary revascularizations (vitamins, 178 [3.0%], vs placebo, 152 [2.5%]; RR, 1.18; 95% CI, 0.95-1.46). Nor were there significant differences in the numbers of deaths attributed to vascular causes (vitamins, 578 [9.6%], vs placebo, 559 [9.3%]) or nonvascular causes (vitamins, 405 [6.7%], vs placebo, 392 [6.5%]) or in the incidence of any cancer (vitamins, 678 [11.2%], vs placebo, 639 [10.6%]).

Conclusion Substantial long-term reductions in blood homocysteine levels with folic acid and vitamin B₁₂ supplementation did not have beneficial effects on vascular outcomes but were also not associated with adverse effects on cancer incidence.

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tenuation of any effects by population-wide folic acid fortification, or lack of real benefit.^{13,14} A subgroup analysis of the HOPE-2 trial^{9,15} and a meta-analysis of a subset of these trials suggested protective effects on stroke.¹⁶ The SEARCH trial assessed the effects of lowering homocysteine levels with folic acid plus vitamin B₁₂ for almost 7

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A list of the SEARCH Collaborative Group members appears in the eAppendix at <http://www.jama.com>.

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years in 12 064 survivors of myocardial infarction.¹⁷ The large size and long duration of treatment in a high-risk population without background mandatory folate fortification allows SEARCH to provide a reliable assessment of the potential benefits and potential hazards of lowering homocysteine substantially with folic acid and vitamin B₁₂ supplementation.

METHODS

The study objectives, design, and methods have been reported previously¹⁷ (protocol appears at <http://www.searchinfo.org>). As well as assessing folate-based homocysteine-lowering therapy, a 2 × 2 factorial design allowed the separate assessment of different doses of simvastatin.

Eligibility

Men and women aged 18 to 80 years with a history of myocardial infarction were eligible provided they had no clear indication for folic acid and had blood cholesterol levels of at least 135 mg/dL if already taking a statin medication or 174 mg/dL if not (to convert cholesterol to mmol/L, multiply by 0.0259). Exclusion criteria are reported elsewhere¹⁷ and included chronic liver, renal, or muscle disease; history of any cancer (except nonmelanoma skin cancer); and use of potentially interacting medications.

Recruitment

Medical collaborators from 88 UK hospitals appointed senior nurses to run special study clinics (see eAppendix, available at <http://www.jama.com>). Approval was obtained from the South East Thames multicenter research ethics committee, and local committees provided site-specific approval. Computerized hospital discharge records were used to identify patients with a diagnosis of myocardial infarction, who, with the agreement of their general practitioners, were invited to attend the local study clinic. At the initial screening visit, a nonfasting blood sample was obtained and guidance given about modification of diet and other vascular

disease risk factors. Eligible individuals were given detailed information about the study and those agreeing provided written consent to participate.

Potentially eligible participants entered a prerandomization run-in phase, which was intended chiefly to limit subsequent randomization to those likely to take the randomly allocated study treatment for several years.¹⁸ Run-in treatment involved placebo vitamin tablets (and 20 mg simvastatin daily, which allowed baseline lipid levels to be assessed after all participants had received the same statin therapy). Individuals who adhered to this treatment, did not have a major vascular event or other serious problem during the run-in, and agreed to participate for several years were randomly allocated to receive 1 tablet daily containing either 2 mg folic acid plus 1 mg vitamin B₁₂ or matching placebo in specially prepared calendar packs (and, separately, using a 2 × 2 factorial design, either 80 mg or 20 mg simvastatin daily). The central telephone randomization system used a minimization algorithm¹⁹ to balance the treatment groups with respect to major prognostic factors.¹⁷

Follow-up

Following randomization between September 1998 and October 2001, participants were to be seen for routine follow-up checks and blood safety monitoring at 2, 4, 8, and 12 months and then every 6 months until final follow-up visits between October 2007 and June 2008. Those who did not attend were to be contacted by telephone at the time of their scheduled follow-up (or alternatively, followed up via their family physician) and could be mailed their allocated study vitamins or placebo tablets (but not their study simvastatin because central blood safety monitoring could not be continued). Adherence was assessed by reviewing the calendar-packed tablets remaining, and for those who had stopped study treatment, the reasons for doing so were sought.

Blood samples were obtained at each follow-up visit for central laboratory assays of alanine aminotransferase and cre-

atine kinase to monitor possible liver- and muscle-related adverse effects of statin therapy. To assess the effects of the vitamin allocation on blood vitamin and homocysteine levels, assays were performed in nonfasting blood collected from a sample of approximately 1000 participants due for follow-up at about the same time each year, from all participants scheduled for follow-up between February 2003 and November 2003 (at median follow-up of 2.5 years) and at all final visits in October 2007 to June 2008. Differences in blood levels between the treatment groups were based on comparisons between all participants allocated vitamins vs all allocated placebo tablets, irrespective of adherence (with any missing data imputed from the baseline values prior to starting any study vitamin treatment).

Information was recorded at each follow-up about any suspected myocardial infarction, stroke, vascular procedure, pulmonary embolus, cancer, or other serious adverse experience, and about the main reasons for all other hospital admissions (including day cases). In addition, reports were systematically sought of muscle pain or weakness and of any serious or nonserious adverse events considered to be probably due to study treatment. Further details were sought from participants' general practitioners (plus, if considered necessary for coding, from any relevant hospital records) about reports that might relate to major vascular events or deaths, and from the Medical Research Information Service for England and Wales and the General Register Office for Scotland about the sites of any registered cancers and the certified causes of any deaths.

All such information was reviewed by coordinating center clinicians who were unaware of the study treatment allocation and events coded according to prespecified criteria.¹⁷ Analyses were based on confirmed plus unrefuted reports, with definite confirmation for 93% of the nonfatal myocardial infarctions, 92% of the nonfatal strokes, and 96% of the revascularizations that were included.

At the final follow-up, cognitive function was assessed using a modified Telephone Interview for Cognitive Status (TICS-m)²⁰ and verbal fluency test,²¹ and hearing thresholds were measured in both ears at 1 KHz and 4 KHz.²²

Statistical Analysis

The data analysis plan was prespecified either in the original protocol or in amendments made before any analyses of the effects of the study treatment on clinical outcomes were available to the steering committee.¹⁷ Comparisons involved log-rank analyses of the first occurrence of particular events during the scheduled treat-

ment period after randomization among all participants allocated active vitamins vs all those allocated placebo (ie, intention-to-treat analyses).^{23,24} The primary comparison was of the effect on the incidence of first major vascular event, defined as nonfatal myocardial infarction or death from coronary heart disease, fatal or nonfatal stroke, or any arterial revascularization.

Secondary comparisons were of the effects on major vascular events in the first year after randomization (when little difference was anticipated) and, separately, in the later years of the treatment period; major vascular events among participants subdivided into 3

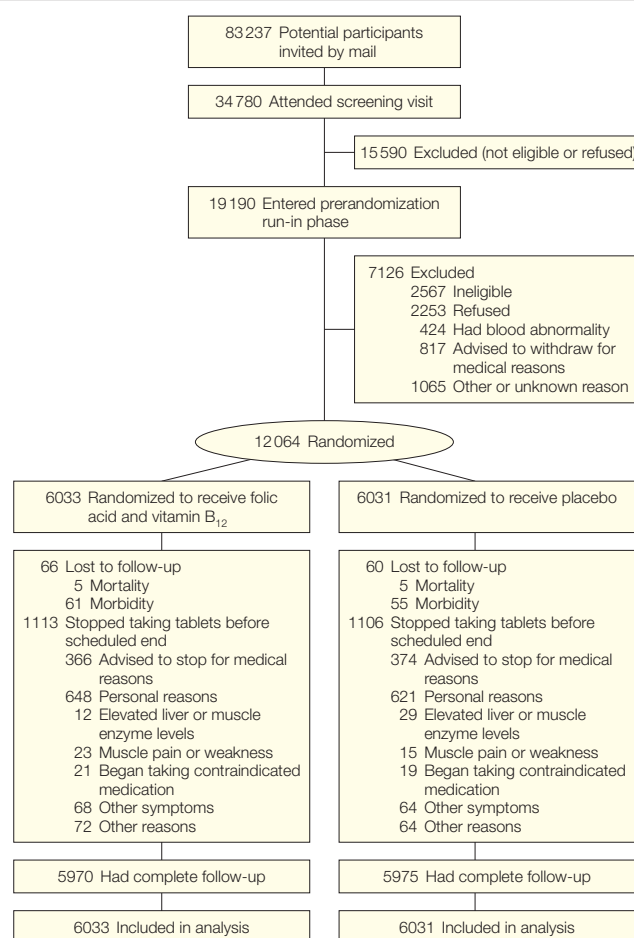
similar-sized groups with respect to blood homocysteine levels at the end of the prerandomization run-in period (before any study vitamin treatment had been taken); major vascular events in the presence of one or other of the allocated study simvastatin regimens; major coronary events, defined as nonfatal myocardial infarction, death from coronary disease, or coronary revascularization; and any type of stroke (excluding transient ischemic attacks).

Tertiary comparisons included the effects on total and cause-specific mortality (considering vascular and nonvascular causes separately), vascular mortality excluding the first year after randomization, coronary and noncoronary revascularization separately, confirmed hemorrhagic and other strokes separately, pulmonary embolus, total and site-specific cancers, hospitalizations for various other causes, and possible adverse effects of treatment. Tests for heterogeneity, or, if more appropriate, trend, were to be used to help determine whether the proportional effects in specific subcategories differed clearly from the overall effects after due allowance for multiple comparisons.^{25,26}

It had been anticipated that allocation to 2 mg folic acid and 1 mg vitamin B₁₂ daily would produce about a 3- to 4- μ mol/L difference in blood homocysteine levels.⁵ (To convert homocysteine to mg/dL, divide by 7.397.) A meta-analysis of observational studies conducted in the early 1990s indicated that a 3 μ mol/L-lower usual homocysteine level was associated with a 20% to 25% lower risk of cardiovascular events.² Based on previous studies among myocardial infarction survivors,^{27,28} it was estimated that approximately 1900 major coronary events would occur during median follow-up of approximately 4 years, which would provide 90% power at 2-sided $P < .01$ to detect a 15% reduction in risk.

It was prespecified in the protocol that the steering committee could modify the study plans while still blinded to the event rates in each treatment group. Following completion of

Figure 1. Flow Diagram for Participants in the SEARCH Trial



Numbers lost to follow-up relate to those without information to the end of the scheduled treatment period on mortality (as well as morbidity) and on morbidity alone. For the reasons given for stopping study tablets, nonstudy statin was started in 92% of the participants who were advised to stop for medical reasons. "Personal reasons" excludes discontinuations attributed to medical reasons. More than 1 reason could be reported.

recruitment, an updated meta-analysis of observational studies reported that a 3 $\mu\text{mol/L}$ —lower usual homocysteine level was associated with only an 11% lower risk of CHD and a 19% lower risk of stroke.³ After median follow-up of 3 to 4 years, both the reduction in risk that might realistically be expected with the homocysteine-lowering therapy (and with the statin comparison^{27,28}) and the overall major coronary event rate were smaller than had been anticipated. Consequently in 2004, blind to interim results for clinical outcomes, the steering committee decided to change the primary outcome from major coronary events to major vascular events and to continue until at least 2800 patients had had a confirmed major vascular event in order to have 90% power at $P < .05$ to detect a 10% reduction in risk.^{17,27,28}

RESULTS

Postal invitations were sent to 83 237 potentially eligible survivors of myocardial infarction and 34 780 attended the initial screening clinic visit, of whom 19 190 remained potentially eligible and entered the prerandomization run-in phase of the study (FIGURE 1). Participants were given a run-in pack containing placebo vitamin tablets (and 20 mg simvastatin daily), and their family physicians were informed of their provisional entry into the trial.

A total of 12 064 individuals (10 012 men and 2052 women) with a history of myocardial infarction were randomized (TABLE), with a mean (SD) age of 64.2 (8.9) years. Previous coronary revascularization was reported by 33%, noncoronary revascularization by 2%, cerebrovascular disease (stroke or transient ischemic attack) by 7%, diabetes mellitus by 11%, and treated hypertension by 42%. At the end of the 2-month prerandomization run-in phase taking placebo-vitamin tablets, the subsequently randomized participants had mean (SD) nonfasting blood levels of homocysteine of 13.5 (4.8) $\mu\text{mol/L}$, folate of 7.4 (4.6) ng/mL, and vitamin B₁₂

of 388 (240) pg/mL. (To convert folate to nmol/L, multiply by 2.266; and vitamin B₁₂ to pmol/L, multiply by 0.7378.) The large size of the trial (and the use of minimized randomization) produced good balance between the treatment groups for the main prerandomization prognostic features that were measured (Table) and should have done likewise for those that were not.

Adherence and Blood Homocysteine

Mean (SD) follow-up duration was 6.7 (1.5) person-years: totals of 40 083 person-years among those allocated vitamins and 40 204 person-years among those allocated placebo. Adherence at each follow-up was defined as at least 80% of the scheduled vitamin or placebo tablets having been taken since the previous follow-up (with only about 2%-3% of participants reporting taking some, but less than 80%, of the treatment). Adherence was similar among those allocated vitamins vs placebo, with 94% in each treatment group adherent after 12 months of follow-up, 89% vs 90% at 48 months, and 84% vs 85% at 84 months.

There were no clear differences between those allocated vitamins or placebo in the proportions reporting different reasons for stopping (Figure 1). Because only a single set of reasons was provided for stopping the study simvastatin, the vitamin tablets, or both, it was not possible to know whether the reason recorded applied to the study vitamin/placebo tablets (but few participants stopped their study vitamins alone). The main reasons given were medical advice and personal wishes, generally because of a perceived need for more intensive low-density lipoprotein-lowering therapy (which is related to the simvastatin comparison).

Compared with placebo, allocation to 2 mg folic acid plus 1 mg vitamin B₁₂ daily reduced blood levels of homocysteine by a mean (SE) of 4.0 (0.3) $\mu\text{mol/L}$ at 12 months, and by 3.3 (0.2) $\mu\text{mol/L}$ at 84 months, yielding a weighted average difference during the study of 3.8 (0.1) $\mu\text{mol/L}$. Folate levels were in-

Table. Baseline Characteristics of Randomized Patients

Characteristic	Patients, No. (%)	
	Vitamins	Placebo
Prior disease		
MI alone	2921 (48)	2924 (48)
Other CHD (+ MI)	2507 (42)	2534 (42)
Other vascular	533 (9)	529 (9)
Diabetes mellitus	635 (11)	632 (10)
Sex		
Male	5006 (83)	5006 (83)
Female	1027 (17)	1025 (17)
Age at randomization, y		
<60	1885 (31)	1880 (31)
≥60 to <70	2414 (40)	2414 (40)
≥70	1734 (29)	1737 (29)
Homocysteine, $\mu\text{mol/L}$		
<11	1735 (29)	1736 (29)
≥11 to <14	2255 (37)	2315 (38)
≥14	2043 (34)	1980 (33)
Folate, ng/mL		
<4.9	1958 (32)	1957 (32)
≥4.9 to <7.9	2108 (35)	2093 (35)
≥7.9	1967 (33)	1981 (33)
Vitamin B ₁₂ , pg/mL		
<298	1829 (30)	1811 (30)
≥298 to <407	2209 (37)	2165 (36)
≥407	1995 (33)	2055 (34)
Total cholesterol, mg/dL		
<155	2488 (41)	2450 (41)
≥155 to <175	1729 (29)	1721 (29)
≥175	1816 (30)	1860 (31)
LDL cholesterol, mg/dL		
<85	1960 (32)	1973 (33)
≥85 to <105	2086 (35)	2091 (35)
≥105	1987 (33)	1967 (33)
Smoking		
Never regular	1372 (23)	1371 (23)
Ex-smoker	3918 (65)	3920 (65)
Current	743 (12)	740 (12)
Estimated GFR (MDRD), mL/min		
<60	852 (14)	834 (14)
≥60	5181 (86)	5197 (86)
Prior treated hypertension		
No	3497 (58)	3493 (58)
Yes	2536 (42)	2538 (42)
Simvastatin allocation		
20 mg daily	3017 (50)	3016 (50)
80 mg daily	3016 (50)	3015 (50)

Abbreviations: CHD, coronary heart disease; GFR (MDRD), glomerular filtration rate estimated using the Modification of Diet in Renal Disease equation; LDL, low-density lipoprotein; MI, myocardial infarction.

SI conversion factors: To convert homocysteine to mg/dL, divide by 7.397; folate to nmol/L, multiply by 2.266; vitamin B₁₂ to pmol/L, multiply by 0.7378; total and LDL cholesterol to mmol/L, multiply by 0.0259.

creased by a weighted mean of 16.2 (0.5) ng/mL and vitamin B₁₂ levels by 625 (19) pg/mL. (No significant differences in plasma lipid levels were seen between these treatment groups.)

Major Vascular Events

During the scheduled treatment period, major vascular events occurred in 1537 of the 6033 participants (25.5%) allocated folic acid plus vitamin B₁₂ vs 1493 of the 6031 (24.8%) allocated placebo (risk ratio [RR], 1.04; 95% confidence interval [CI], 0.97-1.12; *P* = .28) (FIGURE 2). There was no evidence of any benefit beginning to emerge with

more prolonged treatment and follow-up (trend *P* value = .48) (FIGURE 3). Among participants in the low, middle, and high thirds of baseline homocysteine levels, allocation to the study vitamins produced mean (SE) reductions in homocysteine of 2.54 (0.07) μmol/L, 3.34 (0.07) μmol/L, and 5.58 (0.18) μmol/L, respectively, but no significant reduction in major vascular events even among those in the highest third (FIGURE 4). Nor were there any significant differences in the effects on major vascular events in any of the other subcategories of participant studied, including among participants subdivided by random allocation to 80 mg vs 20 mg simvastatin daily.

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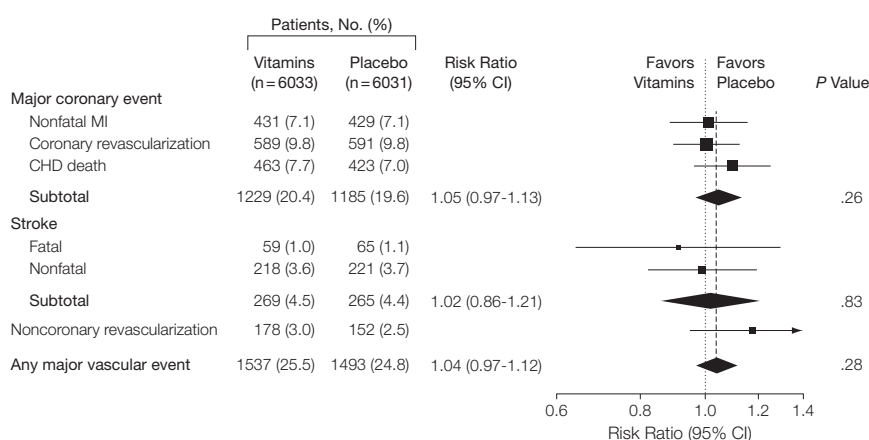
Compared with placebo, allocation to folic acid plus vitamin B₁₂ was associated with nonsignificant adverse trends for death due to coronary heart disease (RR, 1.10; 95% CI, 0.96-1.25) and for noncoronary revascularization (RR, 1.18; 95% CI, 0.95-1.46), but there were no apparent differences for nonfatal myocardial infarction and for coronary revascularization (Figure 2). In addition, allocation to study vitamins was not associated with a significant effect on any stroke (vitamins, 269 [4.5%], vs placebo, 265 [4.4%]; RR, 1.02; 95% CI, 0.86-1.21). Nor were there significant differences in fatal or nonfatal stroke considered separately, or in the numbers of presumed ischemic strokes (vitamins, 243 [4.0%], vs placebo, 245 [4.1%]; RR, 0.99; 95% CI, 0.83-1.19).

For other vascular outcomes, there were no significant differences between the treatment groups in the numbers of patients who were hospitalized for stable or unstable angina (vitamins, 753 [12.5%], vs placebo, 708 [11.7%]), hospitalized or died due to heart failure (vitamins, 260 [4.3%], vs placebo, 248 [4.1%]), or reported to have had transient cerebral ischemic attacks (vitamins, 136 [2.3%], vs placebo, 169 [2.8%]) or any nonfatal or fatal pulmonary emboli (vitamins, 52 [0.9%], vs placebo, 60 [1.0%]).

Mortality

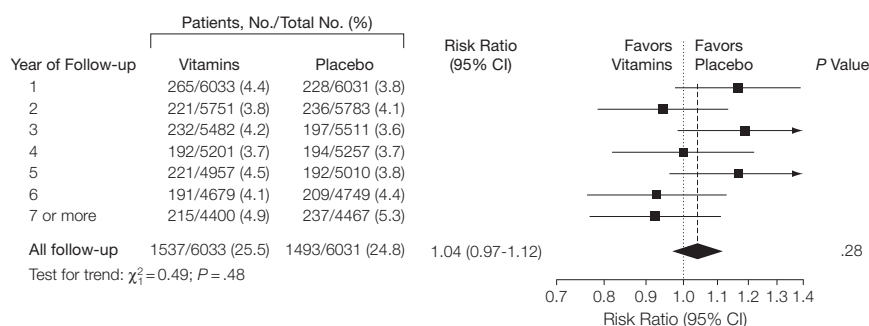
There were no apparent differences in the numbers of deaths attributed to vascular causes (vitamins, 578 [9.6%], vs placebo, 559 [9.3%]; RR, 1.04; 95% CI, 0.92-1.16) or nonvascular causes (vitamins, 405 [6.7%] vs placebo, 392 [6.5%]; RR, 1.04; 95% CI, 0.90-1.19) (FIGURE 5). Among the vascular deaths, allocation to the study vitamins was associated with slightly more deaths due to acute myocardial infarction and coronary heart disease, but those differences were not significant and were partially offset by nonsignificantly fewer deaths from other vascular causes. No significant benefit was seen after ex-

Figure 2. Effects of Folate Allocation on Major Vascular Events: Major Coronary Events, Strokes, and Noncoronary Revascularizations



Analyses are of the numbers of participants having a first event of each type during follow-up (with nonfatal and fatal events considered separately), so there is some nonadditivity between different types of event. Risk ratios compare outcome among participants allocated folic acid and vitamin B₁₂ to that among those allocated placebo. Data markers are approximately proportional to the numbers of events in each subdivision. CI indicates confidence interval; MI, myocardial infarction; CHD, coronary heart disease.

Figure 3. Effects of Folate Allocation on Major Vascular Events by Year of Follow-up



Analyses are of numbers of participants having a first event during each year of follow-up and of those still at risk of a first event at the start of each year. CI indicates confidence interval.

clusion of vascular deaths that occurred in the first year after starting study treatment (vitamins, 8.8%, vs placebo, 8.5%; RR, 1.03; 95% CI, 0.91-1.16). Among the nonvascular deaths, there were no significant differences in the numbers of deaths attributed to neoplastic, respiratory, or other medical or nonmedical causes.

Cancer Incidence

New primary cancers (excluding non-melanoma skin cancer) were diagnosed in 678 (11.2%) of the participants allocated folic acid plus vitamin B₁₂ vs 639 (10.6%) allocated placebo (RR, 1.07; 95% CI, 0.96-1.19) (FIGURE 6) and caused death in 260 (4.3%) vs 252 (4.2%) of the participants (RR, 1.03; 95% CI, 0.87-1.23) (Figure 5). These differences were not significant, nor were there significant differences between the treatment groups in the incidence of cancers in any particular body system (Figure 6).

Other Outcomes

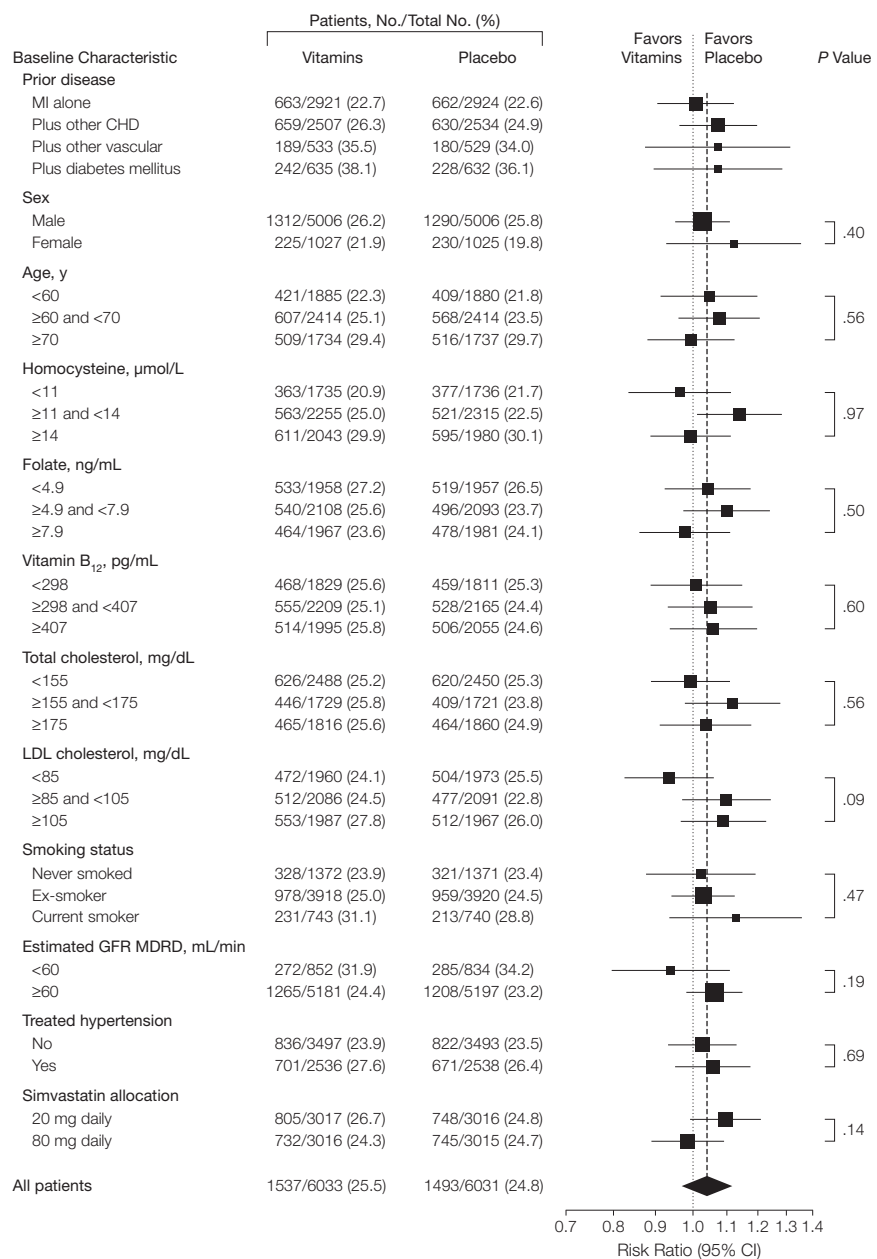
Blood pressure was measured in all participants attending their final follow-up visit, and no differences were detected between the treatment groups. Full blood counts were performed at the 1- and 4-year follow-up visits for each patient, with no significant effects of vitamin allocation seen on any of the hematological parameters measured. However, some conditions known to be associated with deficiencies of these vitamins were less likely to be reported among the participants allocated folic acid and vitamin B₁₂: peripheral neuropathy (vitamins, 7 [0.1%], vs placebo, 17 [0.3%]) and any hematological condition (vitamins, 130 [2.2%], vs placebo, 182 [3.0%]), which includes nonspecific reports of anemia (vitamins, 80 [1.3%], vs placebo, 102 [1.7%]).

Low folate status and increased homocysteine levels have been associated with osteoporosis and fracture risk,³¹⁻³³ but fracture incidence was similar in the 2 groups, both overall (vitamins, 253 [4.2%], vs placebo, 242 [4.0%]) and when only fractures at the

hip, wrist, or spine were considered (vitamins, 85 [1.4%], vs placebo, 69 [1.1%]). Hospitalization rates for all

other outcomes did not differ significantly between the treatment groups (even before making allowance for the

Figure 4. Effects of Folate Allocation on First Major Vascular Event in Different Categories of Participant



P values for χ^2 tests are given for heterogeneity between risk ratios within dichotomous categories and for trend within other categories (except for prior disease categories since there is some overlap between them). Cholesterol categories relate to measured values at the randomization visit after all participants had been taking 20 mg simvastatin daily for 2 months during the prandomization run-in phase. The data analysis plan prespecified that blood homocysteine, vitamin, and cholesterol values would be used to subdivide participants into 3 similar-sized groups. GFR MDRD indicates glomerular filtration rate estimated using the Modification of Diet in Renal Disease equation.^{29,30} CI indicates confidence interval; MI, myocardial infarction; CHD, coronary heart disease; LDL, low-density lipoprotein.

exploratory nature of such analyses), and there were no confirmed reports of serious adverse reactions to the vitamins.

One small randomized trial had reported that folate supplementation might improve age-related hearing loss.³⁴ As would be expected,³⁵ measured hearing thresholds increased with age at both the lower and higher frequencies (with a greater age-related effect at the higher frequency). But at neither frequency was there any significant effect of allocation to folic acid and vitamin B₁₂ (eTable, available at <http://www.jama.com>).

Elevated blood homocysteine levels have also been associated with dementia and cognitive impairment in some, but not all, studies.³⁶⁻⁴⁰ A TICS-m score below 22 out of 39 was prespecified to be indicative of cognitive impairment and, as would be expected, was more common among older individuals and among those with a previous stroke. But despite this discriminatory ability, no significant differences were observed between the treatment groups in the percentages of participants classified as cognitively impaired, either overall (22% allocated vitamins vs 22% allocated placebo; $P < .99$) or in various

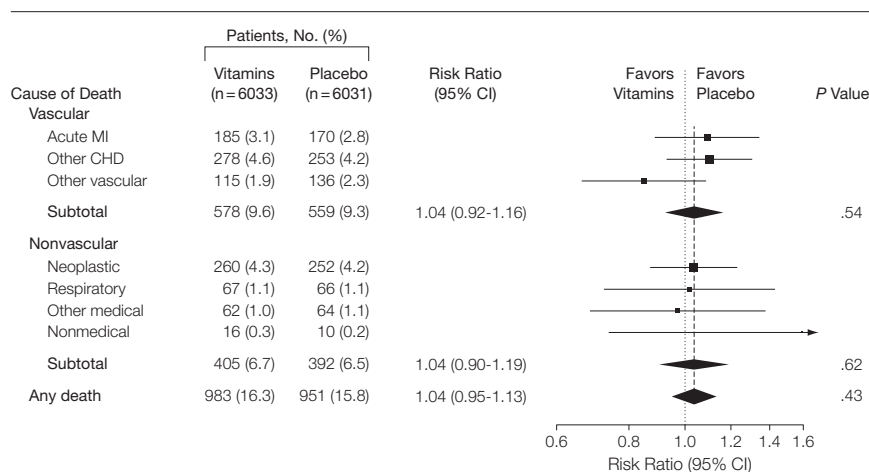
subgroups. Nor was there any significant difference between the groups in mean TICS-m score (24.3 vs 24.3; difference [SE], 0.0 [0.1]) or verbal fluency scores (data not shown). Similar numbers of participants in each treatment group were reported to have developed dementia during follow-up (vitamins, 43 [0.7%], vs placebo, 38 [0.6%]).

COMMENT

The SEARCH trial demonstrates that, despite lowering blood homocysteine levels by 3.8 $\mu\text{mol/L}$ (28%) for up to 7 years in 12 064 high-risk patients, supplementation with 2 mg folic acid and 1 mg vitamin B₁₂ daily had no beneficial effects on major vascular events either overall or in any of the prespecified subgroups studied. Moreover, despite the suggestion that such treatment might preferentially reduce stroke,^{15,16} SEARCH found no significant effects on either strokes or coronary events. A meta-analysis of individual patient data from 8 homocysteine lowering trials (including SEARCH), which has been submitted for publication and includes data on 37 485 individuals, confirms that folic acid supplementation has no significant effects on major vascular events (RR, 1.01; 95% CI, 0.97-1.05) or any of its separate components.

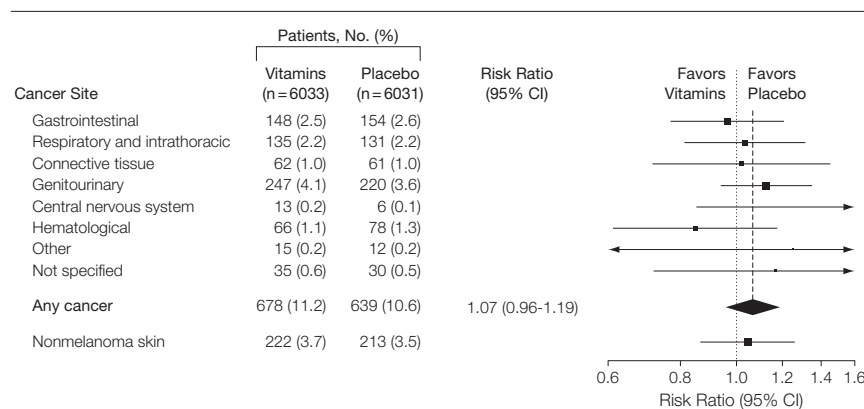
Concerns have been expressed that folic acid may play a role in carcinogenesis, preventing initiation when administered in healthy individuals but potentially promoting tumor growth when administered to individuals with established cancers.⁴¹ It has been suggested that the introduction of folic acid fortification in the United States was linked to an increase in colorectal cancer incidence in the late 1990s.⁴² Previous trials of folic acid supplementation in patients with colorectal polyps have reported conflicting results, with some suggesting possible hazards for those with multiple and advanced colorectal adenomas,⁴³ while others suggested no such effects.^{44,45} In addition, some trials of folic acid supplementation have reported increased risks of

Figure 5. Effects of Folate Allocation on Cause-Specific Mortality



CHD indicates coronary heart disease; CI, confidence interval; MI, myocardial infarction.

Figure 6. Effect of Folate Allocation on Site-Specific Cancer Incidence



Analyses are of the numbers of participants developing cancer at each site (excluding recurrences or new cancers at the same site), so there is some nonadditivity between cancers at different sites. "Connective tissue" excludes nonmelanoma skin cancers, which are given separately. CI indicates confidence interval.

prostate cancer⁴⁶ and, more recently, of lung cancer.⁴⁷

By contrast, with more than 1300 incident cancers during up to 7 years of treatment with 2 mg folic acid and 1 mg vitamin B₁₂ daily, SEARCH provides no evidence of adverse effects on cancer at any particular site including colorectal cancer in 86 patients (1.4%) allocated vitamins vs 91 patients (1.5%) allocated placebo (RR, 0.95; 95% CI, 0.71-1.27); prostate cancer in 159 (2.6%) vs 135 (2.2%) (RR, 1.18; 95% CI, 0.94-1.49); and lung cancer in 116 (1.9%) vs 122 (2.0%) (RR, 0.95; 95% CI, 0.74-1.23). A planned collaborative meta-analysis of all of the large folate-based trials should be able to provide even more reliable evidence about any effects on site-specific cancer. SEARCH did not find that any excess of cancer emerged with prolonged treatment, and continued surveillance of the study participants using UK cancer and mortality registries should allow even longer-term safety of folic acid supplementation to be assessed.

SEARCH was conducted in the United Kingdom during a period without mandatory folate fortification of flour (albeit with increasing voluntary fortification). The daily dose of folic acid given during SEARCH was about 20 times higher than would typically be consumed with mandatory fortification (about 100 µg/d). Hence, these findings from SEARCH are relevant to the public health debate about the safety of mandatory folate fortification for protecting against neural tube defects.^{41,48}

Taken together with the previous homocysteine-lowering trials, the results of SEARCH indicate that folic acid supplementation has no significant adverse effects on cancer or other major health outcomes, even if it also produces no beneficial effects on cardiovascular disease. In addition, these results highlight the importance of focusing on drug treatments (eg, aspirin, statins, and antihypertensive therapy) and lifestyle changes (in particular, stopping smoking and avoiding excessive weight gain) that are of proven benefit, rather than lowering ho-

mocysteine with folic acid-based vitamin supplements, for the prevention of cardiovascular disease.

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